Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation*

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Abstract

We develop an estimation theory on finite-grid path space for stochastic systems whose two-time transition families violate the Chapman–Kolmogorov factorization and therefore admit no generator on the observed state space whose evolution reproduces the given family. The Onsager–Machlup (OM) action, classically a path-density object on tubes (Dürr–Bach, 1978) or the Freidlin–Wentzell rate functional in the small-noise limit, is repurposed as a *selection functional*: the discrete action $S_{\nu,M}[\xi]$, computed against a tractable surrogate model that is explicitly not the truth, acts on candidate paths on a finite time grid through the selection weight $w_\nu(\xi) = \exp(-\lambda S_{\nu,M}[\xi])$. The finite grid is the primary setting, not a truncation: under diffusion proposals the continuous-path OM weight vanishes almost surely, so the continuum formulation is degenerate, and its renormalized theory is deferred to future work. The central claim is methodological: the resulting self-normalized importance-sampling (SNIS) estimator consistently evaluates observables under the path measure selected jointly by the proposal and the surrogate action. Four groups of results are established. (A) Strong consistency, a central limit theorem with the standard self-normalized asymptotic variance, and asymptotic mean-squared error under uniform integrability. (B) Finite-sample concentration in two tiers: polynomial bounds under finite chi-square divergence, via a Cantelli bound on the random denominator, and Bernstein–Hoeffding exponential bounds under bounded weights. (C) Degeneracy diagnostics relating the asymptotic variance and the effective sample size to the chi-square divergence between the selected measure and the proposal, with the Chatterjee–Diaconis and Sanz-Alonso sample-size scales stated as diagnostics under their hypotheses. (D) A conditional transfer to the true process under an explicitly assumed measure-level approximation hypothesis, complemented by an unconditional obstruction: no divisible surrogate family can uniformly approximate an indivisible target's kernels below a deficit-determined threshold. Whether the approximation hypothesis can be controlled by indivisibility invariants—the integrated deficit or the memory rank of the companion projection theory—is an open problem stated in the same form across the program. Three surrogate families are characterized (Ornstein–Uhlenbeck, polynomial drift, and Mori–Zwanzig with explicit memory truncation), and a fully worked one-dimensional example is exhibited in which the selected measure, the estimand, the chi-square divergence, and the asymptotic variance are all available in closed form, with a numerical convergence study against the exact benchmark.

Keywords. Onsager–Machlup functional, selection functional, self-normalized importance sampling, Gibbs reweighting, non-Markovian stochastic processes, Chapman–Kolmogorov violation, effective sample size, concentration inequalities, path-space inference

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Significance statement

This paper provides estimation methodology for stochastic systems whose two-time transition operators violate the Chapman–Kolmogorov relation and therefore admit no generator on the observed state space reproducing them. Because no such generator exists, the classical tools of stochastic modeling—generator-derived likelihoods, generator-derived posteriors on path space, and generator-based rate functionals—are all unavailable on the observed description. We show that the Onsager–Machlup action can nonetheless be repurposed as a *selection functional* that reweights externally proposed candidate paths on a finite time grid against a deliberately imperfect surrogate, yielding a self-normalized importance-sampling estimator with explicit consistency, central-limit, finite-sample concentration, and effective-sample-size guarantees. What the estimator computes is stated exactly: expectations under the path measure selected jointly by the proposal and the surrogate action. The connection to the true process is stated separately, as a conditional transfer under an explicit approximation hypothesis together with an unconditional impossibility result bounding every divisible surrogate away from an indivisible target; closing the gap between the two is identified as the program's open bridge. The framework supplies uncertainty quantification for a class of non-Markovian systems, arising in turbulent transport and metastable molecular dynamics, for which existing methodology assumes structure that does not exist.

1 Introduction

The Onsager–Machlup (OM) action functional has two classical operational roles, both predicated on the existence of a generator. For an Itô stochastic differential equation (SDE)

$$dX_t \;=\; b(X_t)\,dt \;+\; \sigma\,dW_t, \qquad t\in[0,T],\; X_t\in\mathbb{R}^d, \tag{1}$$

with non-degenerate diffusion $\sigma\sigma^\top \succ 0$, the OM action of an absolutely continuous trajectory $\xi\in C([0,T];\mathbb{R}^d)$,

$$S^{\mathrm{OM}}[\xi] \;=\; \frac{1}{2}\int_0^T \left|\sigma^{-1}(\dot\xi - b(\xi))\right|^2 dt + \frac{1}{2}\int_0^T \nabla\cdot b(\xi)\,dt, \tag{2}$$

serves either (i) as the asymptotic relative density of the law of X on ε-tubes around ξ in path space (Dürr–Bach [22]; Ikeda–Watanabe [40]; Capitaine [11]; Pinski–Stuart–Theil [59, 60]; Selk–Honnappa [62]; Ayanbayev–Klebanov–Lie–Sullivan [4]), or (ii) as the Freidlin–Wentzell rate functional governing small-noise large deviations [25, 19, 30]. Recent extensions cover stochastic partial differential equations with Lévy noise [38], jump-diffusions [13, 39], and time-varying noise [74]. In every case the action takes the drift–diffusion pair (b,σ) as an input.

For *indivisible* stochastic processes [5, 6, 50, 64, 65, 66] this input is unavailable. An indivisible target on a configuration space Ω is specified by a two-time column-stochastic kernel family $\{\Gamma(t\leftarrow s)\}_{0\le s\le t\le T}$ that violates the Chapman–Kolmogorov (CK) factorization, in that

$$\Gamma(t\leftarrow s) \;\neq\; \Gamma(t\leftarrow r)\,\Gamma(r\leftarrow s) \qquad \text{for some } s\le r\le t. \tag{3}$$

By Proposition 3.6 of Summers (Paper A) [65], no strongly measurable, Bochner-integrable family of bounded generators reproduces such a family through a time-ordered exponential on $L^2(\Omega,\mu)$; and since an Itô diffusion is Markov while Markov transition kernels satisfy Chapman–Kolmogorov, no drift–diffusion pair (b,σ) on Ω produces (1) with these two-time kernels either, so both classical

roles of (2) collapse simultaneously (Proposition 2.1). We are careful about the scope of this exclusion, which earlier versions of this paper overstated: CK failure of the observed family excludes a generator *on the observed state space* whose evolution reproduces the given two-time family. It does not exclude a Markovian enlargement with hidden variables—indeed, the companion projection theory [67] is built around observed indivisible families arising as compressions of larger flat (divisible) transports—and it does not exclude a path-space law: many non-Markovian processes have perfectly well-defined path measures. The three exclusions (no generator on the observed space; no finite hidden-Markov lift; no Markov lift of any kind) are distinct, and only the first follows from (3).

This paper takes the breakdown as the structural starting point and develops the action in a third role. We treat the discretized form of (2) as a *selection functional* acting on a separately supplied ensemble of candidate paths on a finite time grid, computed against a chosen surrogate model that is explicitly *not* the true dynamics. The proposal mechanism is left abstract: any procedure that generates admissible grid paths and admits i.i.d. sampling is a valid input. The surrogate is a parameterized drift–diffusion pair that defines a quadratic functional on each candidate path. The weighted ensemble average of any path observable against the exponentiated negative surrogate action defines a self-normalized importance-sampling (SNIS) estimator. The central thesis of the paper is the one its theorems prove: *the estimator consistently evaluates observables under the path measure selected jointly by the proposal and the surrogate action.* That selected measure is a well-defined Gibbs reweighting of the proposal; it is not, and is nowhere claimed to be, the law of the target process. The relation to the target is treated separately in §7, as a conditional transfer under an explicitly assumed measure-level approximation hypothesis, together with an unconditional obstruction theorem quantifying why no divisible surrogate, however well chosen, can match the target's kernels.

Why the finite grid is the primary setting The continuous-path formulation of the selection functional is degenerate, not merely inconvenient: under the diffusion proposals natural to this framework (Brownian bridges and Ornstein–Uhlenbeck (OU) processes), sample paths are almost surely not absolutely continuous, so the convention $S_\nu^{\mathrm{OM}}[\xi] = +\infty$ off the absolutely continuous paths forces $w_\nu = 0$ almost surely, the normalizing constant vanishes, and the estimator is $0/0$ (Proposition 2.3). All definitions and theorems of this paper are therefore stated and proved on a fixed finite time grid, where the discrete action is a finite sum, the weights are strictly positive, and the selected measure exists unconditionally. The grid is the model, not a truncation of one; a continuous-path theory (tube-limit renormalization, Cameron–Martin shifts, or a proved discretization limit in the spirit of the Γ-convergence results of [4, 60]) is deferred to future work, and the worked example of §9 exhibits the continuum degeneracy quantitatively as a blow-up of the chi-square divergence under grid refinement.

Application classes The framework applies to any setting in which one has (i) an empirical two-time transition family that fails Chapman–Kolmogorov on the chosen state description and (ii) an external mechanism producing candidate paths. Two concrete classes are: (a) turbulent boundary-layer transitions in geophysical flows, where Mori–Zwanzig memory effects are well documented and divisible Markovian closures are known to fail [46, 47, 28, 63]; and (b) metastable molecular dynamics with insufficient time-scale separation, where the projected coarse-grained dynamics is non-Markovian [32, 55]. The construction is independent of the application class: it requires only the proposal–surrogate pair, and no algorithmic specifics of either are assumed in the analysis.

Contributions The paper makes the following contributions.

(C1) A formal definition of the finite-grid OM selection functional $S_{\nu,M}[\xi]$ acting on candidate grid paths ξ produced by an exogenous proposal Π_M and computed against a surrogate with parameter ν; the associated Gibbs-selected measure $Q_{\nu,M}$ and self-normalized estimator $\hat{F}_N(\nu)$; and a well-posedness lemma showing that, for surrogates with drift divergence bounded below, the weights are bounded and the finite-chi-square condition holds automatically (§3).

(C2) **Theorem A** (§4): strong consistency of $\hat{F}_N(\nu)$ for $F^*(\nu) = \mathbb{E}_{Q_{\nu,M}}[F]$ under minimal moment conditions; a central limit theorem with the standard SNIS asymptotic variance $\sigma^2(\nu) = Z_\nu^{-2}\,\mathbb{E}_{\Pi_M}[(F-F^*)^2 w_\nu^2]$; and convergence of the scaled mean-squared error to $\sigma^2(\nu)$ under uniform integrability. No finite-N mean-squared-error identity is claimed from second moments alone; the random denominator forbids one.

(C3) **Theorem B** (§5): finite-sample concentration in two tiers. Under finite chi-square divergence, polynomial bounds—a Chebyshev bound on the error and a Cantelli bound $\mathbb{P}(B_N < Z_\nu/2) \le 4\chi^2/N$ on the random denominator—together with a genuine finite-N mean-squared-error bound for bounded observables. Under bounded weights, Bernstein and Hoeffding exponential bounds with the correct $W_{\max}^2$ scaling in the denominator tail.

(C4) **Theorem C** (§6): degeneracy diagnostics. The identity $\sigma^2(\nu) = \mathbb{E}_{Q_{\nu,M}}[(F-F^*)^2\rho_\nu]$ for the selected-measure density ρ_ν, the bound $\sigma^2(\nu) \le 4F_{\max}^2(1+\chi^2)$, almost-sure convergence of the plug-in effective sample size to $N/(1+\chi^2)$, and the unconditional ordering $\exp(D_{\mathrm{KL}}) \le 1+\chi^2$. The Chatterjee–Diaconis and Sanz-Alonso sample-size results are stated as diagnostic scales with their hypotheses reproduced.

(C5) **Theorem D** (§7): a conditional transfer to the true process. If the target family is process-induced and the selected measure approximates the true grid law to total-variation tolerance $\eta_{\nu,M}$—an explicit hypothesis, not a derived bound—then the estimator's limit is within $2F_{\max}\,\eta_{\nu,M}$ of the true expectation. Complementing it, an unconditional obstruction (Proposition 7.3): under the standing contraction assumption of [65], every divisible comparison family sits at operator-norm distance at least $D(t,s,r)/3$ from an indivisible target at the worst time pair. Whether $\eta_{\nu,M}$ can be controlled by the integrated deficit or by the memory rank is posed as an open problem, matching its statement in the companion projection theory [67].

(C6) Three surrogate-family characterizations—Ornstein–Uhlenbeck, polynomial drift, and Mori–Zwanzig divisible closure defined by explicit truncation of the memory kernel—together with calibration stated as M-estimation under explicit regularity assumptions (§8).

(C7) A fully worked one-dimensional example (§9) in which the selected measure is Gaussian and every reported quantity—the estimand $F^*(\nu)$, the chi-square divergence, the Kullback–Leibler divergence, and the asymptotic variance $\sigma^2(\nu)$—is evaluated in closed form, with a 50-replication numerical convergence study against the exact benchmark; and a structural comparison (§10) with misspecified-model theory, path-space variational inference, and generalized Bayes, organized around the two structural distinctions that survive scrutiny.

Distinction from prior self-normalized importance sampling The estimator (10) is structurally an SNIS estimator with proposal Π_M and target weight w_ν. Two features distinguish the

construction from prior SNIS theory and from the closely related path-space variational coarse-graining of Harmandaris–Kalligiannaki–Katsoulakis–Plecháč [32]: (i) no generator on the observed state space reproduces the target's two-time family (Proposition 2.1), so $w_\nu \, d\Pi_M$ is not the unnormalized posterior of any generator-based Bayesian model on the observed description; (ii) the surrogate is not optimized toward the truth in any divergence: it is held fixed as a reweighting object whose role is purely computational, and Proposition 7.3 shows that optimizing it toward the target's kernels would in any case hit a deficit-determined floor. The truth link is correspondingly not a theorem of this paper but a conditional transfer under an explicit hypothesis (Theorem 7.1), with the control of that hypothesis by indivisibility invariants left open. §10 makes the comparison to misspecified-model theory, variational inference, and generalized Bayes precise.

Relationship to the program This paper is the estimation-methodology member of a program on indivisible stochastic dynamics. Paper 0 [64] argues on foundational grounds that instantaneous state descriptions are structurally incomplete carriers of dynamical information, motivating the indivisibility framing. Paper A [65] develops the spectral theory of two-time transition families without semigroup structure; the present paper imports from it only four self-contained items, cited at their point of use: the deficit operator $\Delta(t, s, r)$ and its norm $D(t, s, r)$, the no-generator proposition (Proposition 3.6 there), the universal three-pair obstruction bound under a standing contraction assumption (Corollary 4.6 there), and the integrated deficit $J(\mathcal{R})$ with its vanishing characterization (Theorem 7.4 there). Paper B [66] develops convex-perturbation suppression theory for the deficit; it is cited for context only, and no result of the present paper depends on it. The companion projection theory (Paper D) [67] constructs the liftability profile and the memory rank, whose conjectured control of the truth-link tolerance is the shared open bridge (Remark 7.2); the memory-production balance law of [68] quantifies the Mori–Zwanzig memory that the divisible closure of §8.3 explicitly truncates. No dilation-theoretic machinery is used anywhere in this paper.

Causal structure The discrete action is causal: the restriction of $S_{\nu,M}$ to the first m increments depends only on $\xi_0, \ldots, \xi_m$, so the selection weight factorizes across increments (Remark 3.9). No time-reversal symmetry is assumed or used.

Outline §2 recalls the classical OM action, states the no-input proposition with its corrected scope, and proves the continuous-path degeneracy that makes the finite grid primary. §3 formalizes the grid path space, the proposal, the surrogate, the discrete action, the selected measure, and the estimator. §4–§6 establish Theorems A–C. §7 states Theorem D and the obstruction. §8 characterizes the three surrogate families and calibration. §9 works one example in full against exact benchmarks. §10 carries out the structural comparison. §11 states open problems.

2 The classical Onsager–Machlup action and its two breakdowns

We work on a finite horizon $[0, T]$. Let $X = (X_t)_{t\in[0,T]}$ be the Itô diffusion (1) on $\mathbb{R}^d$, with $b \in C_b^1(\mathbb{R}^d; \mathbb{R}^d)$, constant non-degenerate $\sigma \in \mathbb{R}^{d\times d}$, and initial value $X_0 = x_0$. The infinitesimal generator $L = b\cdot\nabla + \frac{1}{2}\sigma\sigma^\top : \nabla^2$ acts on a dense subspace of $C_b(\mathbb{R}^d)$. The law μ_X of X on $C([0, T]; \mathbb{R}^d)$ is equivalent to the law of $x_0 + \sigma W$, with Radon–Nikodym density given by Girsanov's theorem.

Tube density Dürr–Bach [22] and Ikeda–Watanabe [40, Section VI.9] show that for absolutely continuous ξ with $\xi(0)$ fixed,

$$\lim_{\varepsilon\downarrow 0} \frac{\mu_X(\{\eta : \sup_t |\eta_t - \xi_t| \le \varepsilon\})}{\mu_X(\{\eta : \sup_t |\eta_t - \xi^\circ_t| \le \varepsilon\})} \;=\; \exp\bigl(-(S^{\mathrm{OM}}[\xi] - S^{\mathrm{OM}}[\xi^\circ])\bigr). \tag{4}$$

The two ingredients of S^{OM} both encode the generator: b enters through the kinetic term $|\sigma^{-1}(\dot\xi - b)|^2$ and the divergence term $\nabla\cdot b$, which is the Itô correction from the Girsanov density of μ_X relative to the law of $x_0 + \sigma W$.

Freidlin–Wentzell rate functional Replacing $\sigma \to \varepsilon\sigma$ produces the small-noise family X^ε. Freidlin–Wentzell [25] prove the large deviation principle on $C([0,T];\mathbb{R}^d)$ with rate ε^2 and good rate functional $I[\xi] = \frac{1}{2}\int_0^T |\sigma^{-1}(\dot\xi - b(\xi))|^2\,dt$, i.e. the kinetic part of S^{OM} [62]. The minimum-action method [23] and the geometric minimum-action method [34] use this functional as a numerical tool; both start from (b,σ).

2.1 Indivisible targets and the no-input proposition

Following [65], an *indivisible target* is a two-time family $\{\Gamma(t \leftarrow s)\}_{0\le s\le t\le T}$ of column-stochastic kernels on Ω with $\Gamma(s \leftarrow s) = \mathrm{id}$ that satisfies (3). The *deficit operator* at a triple $0 \le s < r < t \le T$ is

$$\Delta(t,s,r) \;=\; \Gamma(t \leftarrow s) \;-\; \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s), \tag{5}$$

with operator norm $D(t,s,r) := \|\Delta(t,s,r)\|_{\mathrm{op}}$ on $L^2(\Omega,\mu)$, under the boundedness conventions of [65, §3]. Indivisibility is $D(t,s,r) > 0$ at some triple. By [65, Proposition 3.6], an indivisible family admits no strongly measurable, Bochner-integrable family of bounded generators reproducing it through a time-ordered exponential.

Proposition 2.1 (No input for the classical roles)**.** *Let $\{\Gamma(t \leftarrow s)\}$ be an indivisible target on $\Omega \subseteq \mathbb{R}^d$. Then there exists no drift–diffusion pair (b,σ) as in* (1) *whose diffusion has two-time transition kernels equal to $\Gamma(t \leftarrow s)$. Consequently neither the tube-density role* (4) *nor the Freidlin–Wentzell rate-functional role of the OM action is available for the target.*

Proof. An Itô diffusion is a Markov process, and the two-time transition kernels of a Markov process satisfy the Chapman–Kolmogorov factorization [65, Proposition 2.6]. A pair (b,σ) reproducing the target's kernels would therefore force $\Delta \equiv 0$, contradicting indivisibility. □

Remark 2.2 (Scope of the exclusion). Proposition 2.1 excludes exactly what its proof uses: a *Markov diffusion on the observed state space* Ω with the given two-time kernels. Three successively stronger exclusions must be kept distinct:

(E1) no generator on the observed state space whose evolution reproduces the given two-time family;

(E2) no finite hidden-Markov enlargement whose compression reproduces the family;

(E3) no Markovian enlargement of any kind.

Only (E1) follows from (3), and it is all this paper claims. (E2) is generically *false*: compressions of finite flat transports are precisely the model sets of the companion projection theory, and every process-induced family on a finite grid admits a finite history lift [67]. (E3) does not follow either.

Likewise, CK failure does not exclude a path-space law: a process-induced indivisible family has a perfectly well-defined law on path space; what fails is the generator-based derivation of a Girsanov-type path density from the observed two-time data. The converse direction is also delicate—two-time CK does not imply Markovianity (Feller's construction [24]; [65, Corollary 2.7])—so the two-time kernels neither pin down the path law nor certify a generator beyond (E1).

2.2 Degeneracy of the continuous-path formulation

The natural continuous-path definition of a selection weight sets, for a surrogate pair (b_ν, σ_ν),

$$w_\nu(\xi) \;=\; \exp\bigl(-\lambda\, S_\nu^{\mathrm{OM}}[\xi]\bigr) \;\text{ for absolutely continuous } \xi, \qquad w_\nu(\xi) \;=\; 0 \;\text{ otherwise,} \tag{6}$$

with S_ν^{OM} the functional (2) evaluated at (b_ν, σ_ν). This formulation is degenerate for exactly the proposals the framework needs.

Proposition 2.3 (Continuous-time weights vanish almost surely)**.** *Let* Π *be any probability measure on* $C([0,T];\mathbb{R}^d)$ *that assigns probability zero to the absolutely continuous paths. Then the weight* (6) *satisfies* $w_\nu = 0$ Π*-almost surely, the normalizing constant* $Z_\nu = \mathbb{E}_\Pi[w_\nu]$ *vanishes, and the self-normalized estimator is undefined* (0/0) *for every sample size. The hypothesis holds for every proposal in the intended class: (i) the law of* σW *for a Brownian motion* W*; (ii) the Brownian bridge; (iii) the Ornstein–Uhlenbeck process from any initial distribution; and (iv) any Itô diffusion* (1) *with non-degenerate constant* σ *whose drift satisfies a Girsanov (Novikov-type) condition—in particular, any bounded drift—so that, conditionally on the initial value* $X_0 = x$*, its law on* $C([0,T];\mathbb{R}^d)$ *is equivalent to that of* $x + \sigma W$*.*

Proof. The conclusions are immediate from (6) once Π assigns probability zero to the absolutely continuous paths, so it suffices to verify the hypothesis in the four cases. (i) Brownian sample paths are almost surely nowhere differentiable, hence not absolutely continuous on any subinterval [52, Theorem 1.30]. (ii) The bridge $B_t = W_t - (t/T)\,W_T$ differs from a Brownian path by the random linear function $t \mapsto (t/T)\,W_T$; adding or subtracting a smooth path preserves absolute continuity in both directions, so the bridge's absolutely continuous paths form a null set by (i). (iii) By the Doob representation, the OU process $dX_t = -k(X_t - \bar{x})\,dt + \sigma\,dW_t$ with $k > 0$ satisfies $X_t - \bar{x} = e^{-kt}\bigl(X_0 - \bar{x}\bigr) + \sigma\, e^{-kt}\,\widetilde{W}\bigl(\tau(t)\bigr)$ with $\tau(t) = (e^{2kt} - 1)/(2k)$ and $\widetilde{W}$ a Brownian motion. Since $t \mapsto e^{-kt}$ is smooth and bounded away from zero on $[0,T]$ and τ is a smooth strictly increasing time change, X is absolutely continuous exactly when $\widetilde{W} \circ \tau$ is, hence exactly when $\widetilde{W}$ is absolutely continuous on $[0, \tau(T)]$—a null event by (i). (iv) The absolutely continuous paths form a Borel set that is invariant under constant shifts and null for the law of σW by (i), hence null for the law of $x + \sigma W$ for every x; conditionally on $X_0 = x$, Girsanov's theorem [40, Chapter IV] makes the diffusion's law equivalent to that of $x + \sigma W$ under the stated drift condition, so the conditional laws—and, integrating over the initial law, the unconditional law—assign the absolutely continuous paths probability zero. □

Proposition 2.3 is not a technical corner case; it invalidates the continuous-path formulation as written, for every proposal in the intended class. The repair is structural: the object on which the estimator actually operates is a finite time grid, where the discrete action is a finite sum and the weights are strictly positive. The remainder of this paper is formulated on the grid, with the grid size M held fixed in every statement and proof; no result invokes an $M \to \infty$ limit, and the only place several grids appear is the comparative refinement study of §9.4, which evaluates the same closed forms at four fixed grids. A renormalized continuous-path theory—through tube probabilities [22, 11], Cameron–Martin shifts, or Γ-convergence of the discrete actions [4, 60]—is

future work (§11), and the worked example of §9 shows quantitatively how the degeneracy re-emerges as chi-square blow-up if one refines the grid at fixed temperature.

3 Finite-grid selection functionals and the estimator

3.1 Grid path space and observables

Fix a time grid $0 = t_0 < t_1 < \cdots < t_M = T$; for notational simplicity we take it uniform with step $\delta t = T/M$, all statements holding verbatim for non-uniform grids. The *grid path space* is

$$\mathcal{X}_M \;:=\; \Omega^{M+1} \;\subseteq\; \mathbb{R}^{d(M+1)}, \qquad \xi = (\xi_0, \xi_1, \ldots, \xi_M),$$

with its Borel σ-algebra. A *path observable* is a measurable function $F : \mathcal{X}_M \to \mathbb{R}$; the *terminal observables* are those of the form $F_\phi(\xi) = \phi(\xi_M)$ for measurable $\phi : \Omega \to \mathbb{R}$. We keep the typing strict throughout: paths are points of $\mathcal{X}_M$; observables are functions on $\mathcal{X}_M$; densities on Ω, where they appear (the initial density ρ_0 below, and the operator statements of §7), are elements of $L^1(\Omega, \mu)$ or $L^2(\Omega, \mu)$ and are never identified with paths or path functionals.

3.2 The proposal mechanism

Definition 3.1 (Proposal mechanism)**.** A *proposal mechanism* is a probability measure Π_M on $\mathcal{X}_M$ together with an i.i.d. sampling rule producing $\xi^{(1)}, \ldots, \xi^{(N)} \sim \Pi_M$.

Assumption 3.2 (Proposal conditions)**.** We impose the following on Π_M, flagging where each is used.

(P1) *Initial condition.* The law of ξ_0 under Π_M is a prescribed initial distribution on Ω: either a density ρ_0 with respect to μ, or a Dirac mass at a point $x_0 \in \Omega$ (as in the worked example). Used for bookkeeping and for Theorem 7.1, where the target law is conditioned on the same initial distribution.

(P2) *Computable density.* Π_M admits a density with respect to a σ-finite reference measure on $\mathcal{X}_M$, available in closed form or evaluable pointwise. Used only for the closed-form expressions of §9; the estimator itself needs only samples.

(P3) *Finite chi-square.* For every ν in a fixed compact $\Theta_0 \subset \Theta$, $\chi^2(Q_{\nu,M} \,\|\, \Pi_M) < \infty$, where $Q_{\nu,M}$ is the selected measure of Definition 3.5. Used for the central limit theorem, the variance theory, and Tier I of Theorem B; strong consistency needs less (Theorem 4.1(i)). By Lemma 3.6, (P3) holds automatically for surrogates whose drift divergence is bounded below.

Condition (P3) is the mathematically minimal requirement under which the SNIS estimator has finite asymptotic variance uniformly over bounded observables [56, 1, 61]; it is strictly stronger than finiteness of the Kullback–Leibler divergence (Theorem 6.1(iii)) and strictly weaker than uniform boundedness of the weight ratio. Sample paths from standard candidate-generation procedures (forward-model simulators, neural-SDE generators, or bridge constructions), read off on the grid, yield proposal mechanisms satisfying Definition 3.1, provided each path is produced by an independent run; within-run particle ensembles from sequential Monte Carlo are dependent through resampling and belong to the adaptive extension discussed in §11.

3.3 The surrogate and the discrete selection action

Definition 3.3 (Surrogate)**.** A *surrogate* is a parameterized family $\{(b_\nu, \sigma_\nu)\}_{\nu\in\Theta}$, $\Theta \subseteq \mathbb{R}^p$ open, with $b_\nu \in C^1(\mathbb{R}^d;\mathbb{R}^d)$ and constant non-degenerate $\sigma_\nu \in \mathbb{R}^{d\times d}$, $\sigma_\nu\sigma_\nu^\top \succeq cI$ with $c > 0$ locally uniformly in ν, and $\nu \mapsto (b_\nu, \sigma_\nu)$ continuous in C^1 on compact sets and in matrix norm. The *discrete selection action* of a grid path $\xi \in \mathcal{X}_M$ at parameter ν is

$$S_{\nu,M}[\xi] \;=\; \frac{1}{2}\sum_{m=0}^{M-1} \frac{\big|\sigma_\nu^{-1}\big(\xi_{m+1} - \xi_m - b_\nu(\xi_m)\,\delta t\big)\big|^2}{\delta t} \;+\; \frac{1}{2}\sum_{m=0}^{M-1} (\nabla\cdot b_\nu)(\xi_m)\,\delta t. \tag{7}$$

Formula (7) is the Euler–Maruyama discretization of the classical functional (2): the kinetic sum discretizes the Girsanov energy and the divergence sum discretizes the Itô correction. On the grid it is a finite sum of finite terms, defined for *every* $\xi \in \mathcal{X}_M$ with no absolute-continuity proviso; this is the entire point of the grid formulation, given Proposition 2.3. No continuum limit of (7) is invoked in this paper.

Remark 3.4 (The surrogate is not the truth)*.* The surrogate is *not* interpreted as an approximation to the target. By Proposition 2.1, no member of any surrogate family is the true dynamics on the observed state space, and Proposition 7.3 bounds every divisible surrogate family away from an indivisible target's kernels by a deficit-determined threshold. The surrogate is a tractable computational object whose only role is to define the functional (7) on candidate paths.

3.4 Selection weight, selected measure, and estimator

Definition 3.5 (Selection weight and selected measure)**.** Fix a temperature $\lambda > 0$. The *selection weight* of $\xi \in \mathcal{X}_M$ at parameter ν is

$$w_\nu(\xi) \;=\; \exp\big(-\lambda\, S_{\nu,M}[\xi]\big) \;\in\; (0,\infty), \tag{8}$$

and, whenever $Z_\nu := \mathbb{E}_{\Pi_M}[w_\nu] \in (0,\infty)$, the *selected measure* is the Gibbs reweighting

$$dQ_{\nu,M} \;=\; \frac{w_\nu}{Z_\nu}\, d\Pi_M, \qquad \rho_\nu := \frac{dQ_{\nu,M}}{d\Pi_M} = \frac{w_\nu}{Z_\nu}. \tag{9}$$

Lemma 3.6 (Well-posedness on the grid)**.** *For every $\xi \in \mathcal{X}_M$ and $\nu \in \Theta$, $0 < w_\nu(\xi) < \infty$. If moreover the drift divergence is bounded below, $(\nabla\cdot b_\nu)(x) \geq -c_\nabla(\nu)$ for all $x \in \Omega$, then*

$$0 \;<\; Z_\nu \;\leq\; W_{\max}(\nu) := \exp\big(\lambda\, c_\nabla(\nu)\, T/2\big) \;<\; \infty,$$

$$\chi^2(Q_{\nu,M} \,\|\, \Pi_M) \;=\; \frac{\mathbb{E}_{\Pi_M}[w_\nu^2]}{Z_\nu^2} - 1 \;\leq\; \frac{W_{\max}(\nu)^2}{Z_\nu^2} - 1 \;<\; \infty,$$

so the selected measure exists and Assumption 3.2(P3) holds automatically. Without the divergence bound, $Z_\nu > 0$ still holds whenever $Z_\nu < \infty$, and (P3) is a genuine assumption.

Proof. $S_{\nu,M}[\xi]$ is a finite sum of finite terms, so $w_\nu(\xi) \in (0,\infty)$ pointwise; positivity of Z_ν follows from $w_\nu > 0$ and Π_M being a probability measure. Under the divergence bound, the kinetic sum in (7) is non-negative and the divergence sum is at least $-c_\nabla T/2$, so $S_{\nu,M} \geq -c_\nabla T/2$ and $w_\nu \leq e^{\lambda c_\nabla T/2}$; bounded weights have all moments, and the displayed chi-square bound is $\mathbb{E}[w^2] \leq W_{\max}^2$. □

For the Ornstein–Uhlenbeck surrogate in dimension d (§8.1), $\nabla \cdot b_\nu = -kd$ is constant, so $W_{\max} = e^{\lambda k T d/2}$ exactly; for polynomial drifts of degree ≥ 2 the divergence is unbounded below and (P3) must be imposed (§8.2). Boundedness of the weights does *not* make the weight distribution benign: the worked example has $W_{\max} = e$ and $W_{\max}/Z_\nu \approx 3.5 \times 10^7$, which is exactly the heavy-tailed-ratio regime the diagnostics of Theorem C are for.

Definition 3.7 (Selection-functional estimator)**.** For a path observable $F : \mathcal{X}_M \to \mathbb{R}$ and an i.i.d. sample $\xi^{(1)}, \ldots, \xi^{(N)} \sim \Pi_M$, the *selection-functional estimator* of F at parameter ν is

$$\hat{F}_N(\nu) \;=\; \frac{\sum_{i=1}^N F(\xi^{(i)})\, w_\nu(\xi^{(i)})}{\sum_{i=1}^N w_\nu(\xi^{(i)})}. \tag{10}$$

The ratio (10) is a self-normalized importance-sampling estimator with proposal Π_M and target $Q_{\nu,M}$, and Theorem A identifies its limit as $F^*(\nu) = \mathbb{E}_{Q_{\nu,M}}[F]$: the estimator consistently evaluates observables under the path measure selected jointly by the proposal and the surrogate action. Under (3) the unnormalized measure $w_\nu\, d\Pi_M$ is *not* the path-space law of any generator-based model on the observed description, and Z_ν has no interpretation as a marginal likelihood; the estimand $F^*(\nu)$ depends on the proposal, the surrogate, and the temperature, and its relation to the target process is exactly the content of §7, no more.

Remark 3.8 (Generalized-Bayes connection)*.* Treating $S_{\nu,M}$ as a path-space loss and λ as a learning rate, the unnormalized measure $w_\nu\, d\Pi_M$ has the form of a Bissiri–Holmes–Walker generalized-Bayes posterior [7, 36, 43], with $\ell = S_{\nu,M}$ acting on a single observation (the candidate path itself). The structural distinction is that ℓ here is not a negative log-likelihood: under indivisibility no generator on the observed state space produces a path-space law whose two-time evolution coincides with the empirical kernels $\Gamma(t \leftarrow s)$, so no generator-derived Girsanov likelihood evaluable at ξ exists on that description. Other path-space measures (those of non-Markovian Gaussian or fractional processes, or of the process inducing the family when one exists) may exist as objects of integration; what fails is the generator-based path-density derivation.

Remark 3.9 (Causal factorization)*.* The action (7) is adapted: its restriction to the first m increments depends only on $(\xi_0, \ldots, \xi_m)$. Consequently the weight factorizes causally, $w_\nu(\xi) = \prod_{m=0}^{M-1} w_\nu^{(m)}(\xi_m, \xi_{m+1})$ with per-increment factors $w_\nu^{(m)} = \exp(-\lambda[\,|\sigma_\nu^{-1}(\xi_{m+1} - \xi_m - b_\nu(\xi_m)\delta t)|^2/(2\delta t) + (\nabla \cdot b_\nu)(\xi_m)\delta t/2\,])$, which is what sequential (particle-filter) implementations exploit.

4 Theorem A: consistency and asymptotic normality

Throughout this section and the next two, Π_M, w_ν, Z_ν, $Q_{\nu,M}$, and $\rho_\nu = w_\nu/Z_\nu$ are as in §3, the parameter $\nu \in \Theta_0$ is fixed, and

$$A_N \;=\; \frac{1}{N}\sum_{i=1}^N F(\xi^{(i)})\, w_\nu(\xi^{(i)}), \qquad B_N \;=\; \frac{1}{N}\sum_{i=1}^N w_\nu(\xi^{(i)}), \qquad \hat{F}_N(\nu) \;=\; \frac{A_N}{B_N}.$$

Theorem 4.1 (A: strong consistency, central limit theorem, asymptotic mean-squared error)**.** *Let $F : \mathcal{X}_M \to \mathbb{R}$ be measurable and suppose $Z_\nu \in (0, \infty)$.*

*(i) (*Strong consistency*) If $\mathbb{E}_{\Pi_M}|F\, w_\nu| < \infty$, then*

$$\hat{F}_N(\nu) \;\xrightarrow{a.s.}\; F^*(\nu) \;:=\; \frac{\mathbb{E}_{\Pi_M}[F\, w_\nu]}{Z_\nu} \;=\; \mathbb{E}_{Q_{\nu,M}}[F] \qquad \text{as } N \to \infty. \tag{11}$$

*(ii) (*Central limit theorem, CLT*) If moreover* $\mathbb{E}_{\Pi_M}[w_\nu^2] < \infty$ *and* $\mathbb{E}_{\Pi_M}[F^2 w_\nu^2] < \infty$*, then*

$$\sqrt{N}\big(\hat{F}_N(\nu) - F^*(\nu)\big) \xrightarrow{d} \mathcal{N}\big(0, \sigma^2(\nu)\big), \qquad \sigma^2(\nu) \;=\; \frac{\mathbb{E}_{\Pi_M}\big[(F - F^*(\nu))^2\, w_\nu^2\big]}{Z_\nu^2}. \tag{12}$$

*(iii) (*Asymptotic mean-squared error*) If, in addition to the hypotheses of (ii), the family* $\{N(\hat{F}_N(\nu) - F^*(\nu))^2\}_{N\geq 1}$ *is uniformly integrable, then*

$$N\,\mathbb{E}\big[(\hat{F}_N(\nu) - F^*(\nu))^2\big] \;\longrightarrow\; \sigma^2(\nu) \qquad \text{as } N \to \infty.$$

Proof. (i) The strong law of large numbers [26] gives $A_N \to \mathbb{E}_{\Pi_M}[F w_\nu]$ and $B_N \to Z_\nu > 0$ almost surely; the continuous mapping theorem applied to the ratio gives (11). (ii) Set $X_i = (F(\xi^{(i)}) - F^*)\, w_\nu(\xi^{(i)})$, which are i.i.d. with $\mathbb{E}X_i = \mathbb{E}_{\Pi_M}[F w_\nu] - F^* Z_\nu = 0$ and $\mathrm{Var}(X_i) = \tilde{\sigma}^2 := \mathbb{E}_{\Pi_M}[(F - F^*)^2 w_\nu^2] < \infty$. Then $\hat{F}_N - F^* = (N^{-1}\sum_i X_i)/B_N$; the classical CLT gives $N^{-1/2}\sum_i X_i \Rightarrow \mathcal{N}(0, \tilde{\sigma}^2)$, $B_N \to Z_\nu$ a.s., and Slutsky's theorem yields (12) with $\sigma^2 = \tilde{\sigma}^2/Z_\nu^2$; this is the standard SNIS asymptotic variance [26, 29, 56], equivalently the delta method applied to $(A_N, B_N) \mapsto A_N/B_N$. (iii) By (ii) and the continuous mapping theorem, $N(\hat{F}_N - F^*)^2 \Rightarrow \sigma^2(\nu)\,\zeta^2$ with $\zeta \sim \mathcal{N}(0,1)$; uniform integrability upgrades convergence in distribution to convergence of expectations. □

Remark 4.2 (No finite-N mean-squared-error identity from second moments). Second moments alone do not yield an exact finite-N bound of the form $\mathbb{E}[(\hat{F}_N - F^*)^2] \leq V/N$: the denominator B_N is random, and on the event that it is small the ratio is uncontrolled by variances. Nor does the asymptotic variance factor as $(1 + \chi^2)\,\mathrm{Var}_{Q_{\nu,M}}(F)$ in general: the correct identity is $\sigma^2(\nu) = \mathbb{E}_{Q_{\nu,M}}[(F - F^*)^2 \rho_\nu]$ (Theorem 6.1(i)), in which the squared deviation and the density ρ_ν are integrated *jointly*, and the product form holds exactly when $(F - F^*)^2$ and ρ_ν are uncorrelated under $Q_{\nu,M}$, which has no reason to hold. Genuine finite-N statements require either the polynomial bounds of Theorem 5.1 (finite chi-square) or the exponential bounds of Theorem 5.2 (bounded weights), both of which control the denominator event explicitly.

5 Theorem B: finite-sample concentration in two tiers

The two tiers are calibrated to the two weight regimes that actually occur (Lemma 3.6): finite chi-square with possibly unbounded weights (Tier I, polynomial concentration), and bounded weights (Tier II, exponential concentration). Finite variance does not imply exponential concentration—an importance weight may have a polynomial tail with finite variance—so the tiers are genuinely distinct.

Theorem 5.1 (B–I: polynomial concentration under finite chi-square)**.** *Assume* $Z_\nu \in (0, \infty)$, $\mathbb{E}_{\Pi_M}|F\, w_\nu| < \infty$ *(so that* $F^*(\nu)$ *is defined),* $\chi^2 := \chi^2(Q_{\nu,M} \| \Pi_M) < \infty$*, and* $\tilde{\sigma}^2 = \mathbb{E}_{\Pi_M}[(F - F^*)^2 w_\nu^2] < \infty$*, and write* $\sigma^2(\nu) = \tilde{\sigma}^2/Z_\nu^2$*. Then for every* $t > 0$ *and every* $N \geq 1$*,*

$$\mathbb{P}\Big(\big|\hat{F}_N(\nu) - F^*(\nu)\big| > t\Big) \;\leq\; \frac{4\,\sigma^2(\nu)}{N\,t^2} \;+\; \frac{4\,\chi^2}{N}, \tag{13}$$

where the second term bounds the denominator event by the Cantelli (one-sided Chebyshev) inequality:

$$\mathbb{P}\Big(B_N < \tfrac{Z_\nu}{2}\Big) \;\leq\; \frac{\mathrm{Var}(w_\nu)/N}{\mathrm{Var}(w_\nu)/N + Z_\nu^2/4} \;\leq\; \frac{4\,\mathrm{Var}(w_\nu)}{N\,Z_\nu^2} \;=\; \frac{4\,\chi^2}{N}, \tag{14}$$

using $\mathrm{Var}(w_\nu) = Z_\nu^2\,\chi^2$. *If moreover* $|F| \le F_{\max}$ Π_M*-a.s., then the finite-*N *mean-squared-error bound*

$$\mathbb{E}\big[(\hat{F}_N(\nu) - F^*(\nu))^2\big] \;\le\; \frac{4\,\sigma^2(\nu)}{N} \;+\; \frac{16\,F_{\max}^2\,\chi^2}{N} \tag{15}$$

holds for every $N \ge 1$.

Proof. Let $X_i = (F(\xi^{(i)}) - F^*)w_\nu(\xi^{(i)})$ as in the proof of Theorem 4.1 and $S_N = \sum_{i=1}^N X_i$, so that on the event $\{B_N \ge Z_\nu/2\}$,

$$\big|\hat{F}_N - F^*\big| \;=\; \frac{|N^{-1}S_N|}{B_N} \;\le\; \frac{2\,|N^{-1}S_N|}{Z_\nu}.$$

Chebyshev's inequality gives $\mathbb{P}(|N^{-1}S_N| > Z_\nu t/2) \le 4\tilde{\sigma}^2/(NZ_\nu^2 t^2) = 4\sigma^2(\nu)/(Nt^2)$. For the denominator, B_N is a mean of i.i.d. variables with mean Z_ν and variance $\mathrm{Var}(w_\nu)/N$; Cantelli's inequality [9, Exercise 2.3] applied to the lower deviation $B_N - Z_\nu \le -Z_\nu/2$ gives (14), and $\mathrm{Var}(w_\nu) = \mathbb{E}[w_\nu^2] - Z_\nu^2 = Z_\nu^2\chi^2$ exactly. The union of the two events gives (13). For (15): $\hat{F}_N$ is a convex combination of values of F, so $|\hat{F}_N| \le F_{\max}$ and $|F^*| \le F_{\max}$ almost surely; splitting the expectation on $\{B_N \ge Z_\nu/2\}$ and its complement,

$$\mathbb{E}\big[(\hat{F}_N - F^*)^2\big] \;\le\; \frac{4}{Z_\nu^2}\,\mathbb{E}\big[(N^{-1}S_N)^2\big] \;+\; (2F_{\max})^2\,\mathbb{P}\Big(B_N < \tfrac{Z_\nu}{2}\Big) \;\le\; \frac{4\tilde{\sigma}^2}{NZ_\nu^2} + \frac{16F_{\max}^2\chi^2}{N}. \qquad \square$$

Theorem 5.2 (B–II: exponential concentration under bounded weights)**.** *Assume additionally* $0 < w_\nu \le W_{\max}$ *and* $|F| \le F_{\max}$ Π_M*-a.s. Then for every* $t > 0$ *and every* $N \ge 1$,

$$\mathbb{P}\Big(\big|\hat{F}_N(\nu) - F^*(\nu)\big| > t\Big) \;\le\; 2\exp\!\left(-\frac{N\,t^2}{8\,\sigma^2(\nu) \;+\; \frac{8}{3}\,F_{\max}W_{\max}\,t/Z_\nu}\right) \;+\; \exp\!\left(-\frac{N\,Z_\nu^2}{2\,W_{\max}^2}\right), \tag{16}$$

the first term by Bernstein's inequality on the numerator sum and the second by Hoeffding's inequality on the denominator, with the $W_{\max}^2$ *scaling dictated by the range of a single weight.*

Proof. The summands $X_i = (F_i - F^*)w_i$ satisfy $|X_i| \le 2F_{\max}W_{\max} =: b$ and $\mathrm{Var}(X_i) = \tilde{\sigma}^2$. Bernstein's inequality [9, Theorem 2.10] gives, for $u > 0$,

$$\mathbb{P}\big(|S_N| \ge Nu\big) \;\le\; 2\exp\!\left(-\frac{Nu^2}{2\tilde{\sigma}^2 + \frac{2}{3}\,b\,u}\right).$$

Taking $u = Z_\nu t/2$ and simplifying with $\tilde{\sigma}^2 = Z_\nu^2\sigma^2(\nu)$ yields the first term of (16) on the event $\{B_N \ge Z_\nu/2\}$. For the denominator, $w_\nu \in [0, W_{\max}]$ has range at most $W_{\max}$, so the one-sided Hoeffding inequality [9, Theorem 2.8] gives

$$\mathbb{P}\Big(B_N - Z_\nu \le -\tfrac{Z_\nu}{2}\Big) \;\le\; \exp\!\left(-\frac{2N(Z_\nu/2)^2}{W_{\max}^2}\right) \;=\; \exp\!\left(-\frac{NZ_\nu^2}{2W_{\max}^2}\right). \qquad \square$$

Corollary 5.3 (Hoeffding form)**.** *Under the hypotheses of Theorem 5.2, replacing Bernstein by Hoeffding on the numerator (summand range* $4F_{\max}W_{\max}$*),*

$$\mathbb{P}\Big(\big|\hat{F}_N(\nu) - F^*(\nu)\big| > t\Big) \;\le\; 2\exp\!\left(-\frac{N\,t^2\,Z_\nu^2}{32\,F_{\max}^2\,W_{\max}^2}\right) \;+\; \exp\!\left(-\frac{N\,Z_\nu^2}{2\,W_{\max}^2}\right). \tag{17}$$

Remark 5.4 (Which tier binds). Tier II applies whenever the divergence bound of Lemma 3.6 holds, but its constants degrade with the ratio $W_{\max}/Z_\nu$: in the worked example, $W_{\max} = e$ while $Z_\nu \approx 7.9\times10^{-8}$, so the exponents in (16)–(17) are vacuous at practical N and the polynomial Tier I together with the chi-square diagnostics of Theorem C carries the operational content. Conversely, when the drift divergence is unbounded below, the automatic bound of Lemma 3.6 is unavailable; unless the proposal itself confines the paths, so that the Π_M-essential supremum of w_ν is finite, Tier I is the only finite-sample statement. Weights with sub-exponential (rather than bounded) tails admit Bernstein-type bounds with the corresponding variance and scale parameters [9, §2.8]; we do not state them separately. The general theory of self-normalized processes [58] provides exponential bounds for ratios of the form $S_N/\sqrt{\sum_i X_i^2}$ under its own hypotheses (symmetry or moment conditions on the self-normalized ratio); those hypotheses are not implied by finite chi-square, and we cite that theory for context rather than invoking it.

6 Theorem C: degeneracy diagnostics and sample-size scales

Theorem 6.1 (C: variance, effective sample size, and divergence ordering)**.** *Assume $Z_\nu \in (0,\infty)$ and $\chi^2 = \chi^2(Q_{\nu,M}\|\Pi_M) < \infty$, and let $\rho_\nu = w_\nu/Z_\nu$ be the selected-measure density* (9)*.*

*(i) (*Variance identity and bound*) The asymptotic variance of* (12) *satisfies*

$$\sigma^2(\nu) \;=\; \mathbb{E}_{Q_{\nu,M}}\big[(F - F^*(\nu))^2\,\rho_\nu\big], \qquad \mathbb{E}_{Q_{\nu,M}}[\rho_\nu] \;=\; 1+\chi^2,$$

and hence $\sigma^2(\nu) \le 4F_{\max}^2\,(1+\chi^2)$ whenever $|F| \le F_{\max}$ Π_M-a.s.

*(ii) (*Effective sample size*) The plug-in effective sample size $\hat N_{\text{eff}} = \big(\sum_i w_\nu^{(i)}\big)^2 / \sum_i \big(w_\nu^{(i)}\big)^2$ satisfies*

$$\frac{\hat N_{\text{eff}}}{N} \;\xrightarrow{\;a.s.\;}\; \frac{1}{1+\chi^2} \qquad \text{as } N\to\infty.$$

*(iii) (*Divergence ordering*) Unconditionally,*

$$\exp\big(D_{\mathrm{KL}}(Q_{\nu,M}\,\|\,\Pi_M)\big) \;\le\; 1+\chi^2.$$

Proof. (i) $\sigma^2 = Z_\nu^{-2}\mathbb{E}_{\Pi_M}[(F-F^*)^2 w_\nu^2] = \mathbb{E}_{\Pi_M}[(F-F^*)^2\rho_\nu^2] = \mathbb{E}_{Q_{\nu,M}}[(F-F^*)^2\rho_\nu]$, and $\mathbb{E}_Q[\rho_\nu] = \mathbb{E}_{\Pi_M}[\rho_\nu^2] = \mathbb{E}[w^2]/Z^2 = 1+\chi^2$; the bound uses $|F-F^*| \le 2F_{\max}$. (ii) By the strong law, $B_N \to Z_\nu$ and $N^{-1}\sum_i w_i^2 \to \mathbb{E}[w^2]$ a.s.; hence $\hat N_{\text{eff}}/N = B_N^2/(N^{-1}\sum_i w_i^2) \to Z_\nu^2/\mathbb{E}[w^2] = 1/(1+\chi^2)$. (iii) By Jensen's inequality for the concave logarithm, $D_{\mathrm{KL}}(Q\|\Pi) = \mathbb{E}_Q[\log\rho_\nu] \le \log\mathbb{E}_Q[\rho_\nu] = \log(1+\chi^2)$; this is the monotonicity of Rényi divergences in the order, $D_1 \le D_2$ [70]. □

Part (i) makes precise in what sense $1+\chi^2$ is the variance-inflation factor of the selection step: it is an upper bound, attained in order of magnitude when the deviation $(F-F^*)^2$ correlates with the density ρ_ν, and the popular reading $N_{\text{eff}} = N/(1+\chi^2)$ [48, 56] is the corresponding worst-case deflator, not an identity for a fixed observable (Remark 4.2). Part (ii) is the sense in which the plug-in diagnostic is consistent; its finite-sample behavior is another matter—the plug-in $\hat\chi^2$ and $\hat N_{\text{eff}}$ are themselves dominated by the upper tail of w_ν and are severely negatively biased exactly in the heavy-tailed regime, a bias the worked example measures against the closed form (§9.4).

Sample-size scales as diagnostics Two published sample-size results attach to the divergences in Theorem 6.1(iii), and we state them as *diagnostic scales*, with their hypotheses, rather than as universal necessary conditions for a fixed observable. Chatterjee–Diaconis [14] show that, for importance sampling of $\mathbb{E}_Q[F]$ with $\|F\|_{L^2(Q)} \le 1$, sample sizes $N \ge \exp(D_{\mathrm{KL}} + t)$ suffice for accuracy with failure probability of order $e^{-t/4}$, while for $N \le \exp(D_{\mathrm{KL}} - t)$ there *exist* bounded observables for which the estimator fails; the scale $\exp(D_{\mathrm{KL}})$ is thus sufficient uniformly over an $L^2(Q)$ ball and necessary for the class, not for each fixed F. Sanz-Alonso [61] shows that guarantees on the relative mean error uniformly over test-function classes require N at the scale $1+\chi^2$—again a class statement, with the constants depending on the class and the accuracy. The two scales are ordered by Theorem 6.1(iii), $\exp(D_{\mathrm{KL}}) \le 1+\chi^2$, and they respond differently to the accuracy level, so neither dominates operationally in all regimes: when the weight ratio is heavy-tailed—χ^2 large relative to $\exp(D_{\mathrm{KL}})$, which is the regime in which a reweighting object is needed in the first place—the chi-square scale is the binding diagnostic, while $\exp(D_{\mathrm{KL}})$ remains the available reference when only the Kullback–Leibler divergence is computable. The worked example sits squarely in the chi-square-binding regime ($\exp(D_{\mathrm{KL}}) \approx 129$ against $1+\chi^2 \approx 1373$; §9.4).

Remark 6.2 (Diagnostics in practice). $\hat{N}_{\mathrm{eff}}$ and the plug-in $\hat{\chi}^2$ are computable at no additional cost during the SNIS sweep. When $\hat{N}_{\mathrm{eff}} \lesssim 10^2$ the estimator is operationally unreliable for all but the most stable observables; when $\hat{N}_{\mathrm{eff}}/N \lesssim 10^{-2}$, Pareto-smoothed importance sampling [71] is recommended as a tail-stabilizer, and the negative bias of the plug-in diagnostics (Theorem 6.1(ii) describes only their limit) should be assumed rather than ruled out.

7 Theorem D: the truth link as a conditional transfer, and the obstruction

Everything so far concerns the selected measure $Q_{\nu,M}$. This section states exactly what can be said about the target process: a conditional transfer under an explicit measure-level approximation hypothesis (Theorem 7.1), an obstruction theorem explaining why the hypothesis cannot be discharged by improving the surrogate (Proposition 7.3), and the open problem that separates them (Remark 7.2).

7.1 Process-induced targets and the grid law

Let $\{\Gamma(t \leftarrow s)\}$ be a two-time family on Ω as in §2.1, and restrict it to the grid $\{t_0, \dots, t_M\}$. Call the restricted family *process-induced* if there exists a stochastic process $(X_{t_m})_{m=0}^M$ on Ω whose conditional transition kernels on the grid are the $\Gamma(t_j \leftarrow t_i)$; for finite Ω, the companion projection theory characterizes exactly this class and produces explicit finite hidden-Markov realizations of it [67]. A process-induced family, together with the initial distribution of Assumption 3.2(P1), determines the law $P_{\mathrm{true},M}$ of $(X_{t_0}, \dots, X_{t_M})$ on $\mathcal{X}_M$ only partially—two-time kernels do not pin down the joint law (Remark 2.2)—so we take $P_{\mathrm{true},M}$ to denote the law of the inducing process itself, fixed once and for all, and write $F_{\mathrm{true}} = \mathbb{E}_{P_{\mathrm{true},M}}[F]$.

Theorem 7.1 (D: conditional transfer)**.** *Let the target be process-induced with grid law $P_{\mathrm{true},M}$, let $F : \mathcal{X}_M \to \mathbb{R}$ satisfy $|F| \le F_{\max}$, and let ν be such that $Z_\nu \in (0,\infty)$. Assume the* measure-level approximation hypothesis

(D1) $d_{\mathrm{TV}}\big(Q_{\nu,M},\, P_{\mathrm{true},M}\big) \;\le\; \eta_{\nu,M}$ *for a known tolerance $\eta_{\nu,M} \ge 0$,*

where $d_{\mathrm{TV}}(P,Q) = \sup_A |P(A) - Q(A)|$. *Then*

$$\left|F^*(\nu) - F_{\mathrm{true}}\right| \;\le\; 2\,F_{\max}\,\eta_{\nu,M}, \tag{18}$$

and combining with Theorems A and B, $\hat{F}_N(\nu) \to F^*(\nu)$ *almost surely with the stated finite-sample deviations, so that the total error decomposes as*

$$\left|\hat{F}_N(\nu) - F_{\mathrm{true}}\right| \;\le\; \underbrace{\left|\hat{F}_N(\nu) - F^*(\nu)\right|}_{\text{estimator error: Theorems A–C}} \;+\; \underbrace{2\,F_{\max}\,\eta_{\nu,M}}_{\text{selection bias: hypothesis (D1)}}.$$

For terminal observables $F_\phi(\xi) = \phi(\xi_M)$ *with* $|\phi| \le F_{\max}$*, (D1) may be weakened to a total-variation bound on the time-*t_M *marginals alone.*

Proof. $|F^*(\nu) - F_{\mathrm{true}}| = |\int F\, d(Q_{\nu,M} - P_{\mathrm{true},M})| \le F_{\max}\,\|Q_{\nu,M} - P_{\mathrm{true},M}\|_{\mathrm{TV}} = 2F_{\max}\, d_{\mathrm{TV}}(Q_{\nu,M}, P_{\mathrm{true},M})$, and for terminal observables the signed measure may be replaced by the difference of the terminal marginals. □

Theorem 7.1 is stated in the form it deserves: the operator-theoretic content of the program controls nothing in it. Hypothesis (D1) is an assumption about the pair (proposal, surrogate) relative to the target, to be established—or estimated—by whatever means the application affords; the theorem's content is the clean split of the total error into an estimator part, which Theorems A–C control unconditionally, and a selection-bias part, which is exactly as good as the hypothesis. An earlier version of this paper claimed an unconditional bound on the selection bias in terms of the integrated deficit of the target family; that claim was withdrawn, and its failure is instructive: when the target is divisible, any such bound would force zero bias for *every* admissible proposal and surrogate, which is absurd, since the selected measure can be made arbitrarily far from a divisible target by choosing a poor proposal.

Remark 7.2 (The open bridge). Whether $\eta_{\nu,M}$ in (D1) can be *controlled* by indivisibility invariants of the target—the integrated deficit $J(\mathcal{R}) = \iiint_{s<r<t} D(t,s,r)^2\, ds\, dr\, dt$ of [65, Theorem 7.4], or the liftability profile h_d and memory rank of [67]—for principled choices of proposal and surrogate, is open. This is the same open problem, in the same form, as the selection bridge conjectured in [67, Open Problem (i)]: for a process-induced two-time family V (in the notation of [67]), the selection-functional readout of V should agree with the readout of the compression of its projection onto the d-state model set, up to an error controlled by $h_d(V)$. The present paper adds the estimator-side half of that program (Theorems A–C) and the precise statement of what remains; it does not close the bridge, and no result above depends on it.

7.2 The obstruction: divisible surrogates cannot match indivisible targets

The conditional structure of Theorem 7.1 is forced, not chosen: no choice of divisible surrogate drives the selection bias to zero, because divisible families are uniformly bounded away from an indivisible target's kernels.

Proposition 7.3 (Deficit obstruction to divisible approximation)**.** *Let* $\{\Gamma(t \leftarrow s)\}$ *be an indivisible target whose two-time operator family satisfies the standing contraction assumption of [65, Assumption 3.3] on* $L^2(\Omega,\mu)$*, and let* $\{\tilde{\Gamma}(t \leftarrow s)\}$ *be* any *divisible two-time family on* Ω *satisfying the same assumption—in particular the family* $\tilde{\Gamma}(t \leftarrow s) = e^{(t-s)L_\nu}$ *induced by any surrogate pair* (b_ν, σ_ν) *whose semigroup is a contraction on* $L^2(\Omega,\mu)$*. Then for every triple* $0 \le s < r < t \le T$*,*

$$D(t,s,r) \;\le\; \delta(t,s) + \delta(t,r) + \delta(r,s), \qquad \delta(\tau_1,\tau_2) := \|\Gamma(\tau_1 \leftarrow \tau_2) - \tilde{\Gamma}(\tau_1 \leftarrow \tau_2)\|_{\mathrm{op}}, \tag{19}$$

and consequently, at every triple with $D(t,s,r) > 0$,

$$\sup_{(\tau_1,\tau_2)} \|\Gamma(\tau_1 \leftarrow \tau_2) - \tilde{\Gamma}(\tau_1 \leftarrow \tau_2)\|_{\mathrm{op}} \;\geq\; \frac{D(t,s,r)}{3}.$$

Proof. This is the universal three-pair bound of [65, Corollary 4.6]: insert $\tilde{\Gamma}(t \leftarrow s) = \tilde{\Gamma}(t \leftarrow r)\tilde{\Gamma}(r \leftarrow s)$ into the deficit by the triangle inequality and bound the composite terms using the contraction hypothesis on one factor of each product. □

Proposition 7.3 formalizes Remark 3.4 at the level of the whole family: no divisible surrogate reproduces an indivisible target's kernels uniformly within $D(t,s,r)/3$ at the worst time pair, for any triple with positive deficit. Two comments on scope. First, the statement is a *family-norm* obstruction—the supremum over time pairs is essential; a divisible family may match $\Gamma(t \leftarrow s)$ exactly at one designated pair while paying at the others, so no single-pair lower bound of this form holds, and none is claimed. Second, the bound is an impossibility result, not an error formula: it bounds surrogate families away from the target's *kernels* in operator norm and says nothing quantitative about the total-variation distance in (D1), which involves the proposal and the temperature as well. The metric refinement of (19) is the dimension-matched charge $h_n(V) = \mathrm{dist}(V, \mathcal{F}\mathrm{lat})$—the $d = n$ member of the liftability profile h_d—of the companion projection theory, two-sidedly comparable to the deficit norms through a Łojasiewicz inequality [67].

8 Surrogate characterizations

We characterize three canonical surrogate families and the calibration of ν from data.

8.1 Ornstein–Uhlenbeck surrogate

Take $b_\nu(x) = -k(x - \bar{x})$ with $\nu = (k, \bar{x}, \sigma) \in (0,\infty) \times \mathbb{R}^d \times \mathrm{Sym}_d^+$ and $\sigma_\nu = \sigma$ constant. The divergence is constant, $\nabla \cdot b_\nu = -kd$, so Lemma 3.6 applies with $W_{\max} = e^{\lambda k T d/2}$ and the selected measure exists with finite chi-square for every proposal. The discrete action (7) is

$$S_{\nu,M}[\xi] \;=\; \frac{1}{2}\sum_{m=0}^{M-1} \frac{\left|\sigma^{-1}\big(\xi_{m+1} - \xi_m + k(\xi_m - \bar{x})\,\delta t\big)\right|^2}{\delta t} \;-\; \frac{kTd}{2}, \tag{20}$$

a quadratic functional of ξ, and the weight factorizes across increments into one-step Gaussian-type factors (Remark 3.9); when the proposal is itself Gaussian on the grid, the selected measure is Gaussian with explicitly computable precision (§9). The restoring rate k is the surrogate's single tunable timescale. We record its correct interpretation: the action accumulates over the entire horizon $[0,T]$—no window truncation is present in (20)—and what k controls is an effective decay of *sensitivity*, the influence of an increment on the surrogate's implied conditional mean decaying on the timescale $1/k$; the functional itself is not local in time.

8.2 Polynomial-drift surrogate

Take $b_\nu(x) = \sum_{|\alpha| \leq M_p} c_\alpha\, x^\alpha$ for multi-indices $\alpha \in \mathbb{N}_0^d$ of degree at most M_p, and $\sigma_\nu = \sigma$. The selection weight is a polynomial-quadratic functional of ξ. For $M_p = 1$ with $b_\nu(x) = Ax + a$, $A \prec 0$, this recovers the multivariate OU surrogate with restoring matrix $K = -A \succ 0$ and $\bar{x} = -A^{-1}a$ (the scalar rate k of §8.1 corresponding to $K = kI$); $M_p \geq 2$ admits double-well and metastable structure. For $M_p \geq 2$ the divergence $\nabla \cdot b_\nu$ is in general a nonconstant polynomial and can be

unbounded below—as for the double-well drift $b(x) = x - x^3$ on $\mathbb{R}$, with $\nabla \cdot b = 1 - 3x^2$; in that case the weights are unbounded, Lemma 3.6's automatic bound is unavailable, and Assumption 3.2(P3) must be verified for the proposal at hand, with the finite-sample statements coming from Tier I of Theorem B.

8.3 Mori–Zwanzig surrogate by explicit memory truncation

A more principled surrogate derives from the Mori–Zwanzig (MZ) formalism [51, 77, 78, 16, 28, 63, 46, 47]. Let P be a projection onto a subspace of resolved observables. The MZ identity decomposes the projected evolution into three parts: a Markovian term generated by the projected dynamics, a memory integral against a kernel supported on the past, and a fluctuating force orthogonal to the resolved subspace. We emphasize what the formalism actually says: *projection produces memory; it does not remove it.* Eliminating degrees of freedom is where the memory integral comes from, and the projected dynamics is generically non-Markovian; no divisible kernel arises from projection alone. A divisible surrogate is obtained from the MZ hierarchy only by a *deliberate closure*: truncate the memory integral (and drop the fluctuating force), retaining the Markovian term, whose drift and diffusion coefficients then define $(b_{\nu^P}, \sigma_{\nu^P})$. The truncation is an additional approximation and is stated as one; its size is precisely the memory the closure discards, and the balance law of the companion work [68] quantifies the corresponding production of Chapman–Kolmogorov failure at the two-time-family level. Proposition 7.3 then bounds how well *any* such divisible closure—MZ-derived or otherwise—can track an indivisible target: at every triple with $D(t, s, r) > 0$, the closure's two-time family is at least $D(t, s, r)/3$ away in operator norm at the worst time pair. The MZ surrogate is therefore exactly what Remark 3.4 says every surrogate is: a principled reweighting object, not an approximation scheme that converges to the truth.

8.4 Calibration as M-estimation

When observed sample paths $\xi_{\mathrm{obs}}^{(1)}, \ldots, \xi_{\mathrm{obs}}^{(n)}$ are available, ν may be calibrated by minimizing the empirical surrogate action,

$$\hat{\nu} \;=\; \arg\min_{\nu \in \Theta_0} \; \frac{1}{n} \sum_{i=1}^{n} S_{\nu,M}\big[\xi_{\mathrm{obs}}^{(i)}\big], \tag{21}$$

an M-estimation proposal in the sense of [37, 75]. We state it as a proposal with its regularity assumptions listed, since the classical consistency and sandwich-covariance theorems do not apply automatically: they require (a) observations that are i.i.d. or satisfy suitable mixing conditions—observed paths may be serially dependent, and a single long realization places the entire burden on the mixing hypotheses; (b) identification, i.e. a unique population minimizer $\nu^* = \arg\min_\nu \mathbb{E}\, S_{\nu,M}[\xi_{\mathrm{obs}}]$ on the compact Θ_0; (c) domination and twice continuous differentiability of $\nu \mapsto S_{\nu,M}$ in a neighborhood of ν^*; (d) a non-singular Hessian at ν^*; and (e) a uniform law of large numbers over Θ_0. Under (a)–(e), $\hat{\nu} \to \nu^*$ and $\sqrt{n}(\hat{\nu} - \nu^*)$ is asymptotically normal with sandwich covariance [75, 69, 53]. The structural caveat survives every regularity regime: ν^* is the best reweighting parameter within the chosen family, with *no* interpretation as the Kullback–Leibler projection of a true generator-based law onto the surrogate family, because no such law exists on the observed description (Proposition 2.1).

9 Worked example: a one-dimensional OU selection functional

We exhibit the full machinery on a one-dimensional example in which every reported quantity is available in closed form. The proposal is a discretized Ornstein–Uhlenbeck process started deterministically at the origin; the surrogate is an Ornstein–Uhlenbeck pair at a faster restoring rate; the observable is nonlinear at the terminal time.

9.1 Setup

Set $\Omega = \mathbb{R}$, $T = 1$, and fix the grid $t_m = m\,\delta t$ with $M = 50$ and $\delta t = T/M = 0.02$. The proposal is the Euler–Maruyama discretization of the OU process with restoring rate $k_\Pi = 1$ and noise $\sigma_\Pi = 1$, *started deterministically at* $\xi_0 = 0$:

$$\xi_{m+1} \;=\; a_\Pi\,\xi_m + \sigma_\Pi\sqrt{\delta t}\,\eta_m, \qquad a_\Pi = 1 - k_\Pi\,\delta t, \qquad \eta_m \sim \mathcal{N}(0,1) \text{ i.i.d.}$$

The deterministic start matters, and we do not describe this proposal as stationary: under Π_M the terminal variance is

$$\mathrm{Var}_{\Pi_M}(\xi_M) \;=\; \delta t\,\frac{1 - a_\Pi^{2M}}{1 - a_\Pi^2} \;=\; 0.438071,$$

the discrete analogue of the continuous-time value $(1-e^{-2})/2 \approx 0.432332$ for an OU process released from the origin, and well below the stationary variance $\frac{1}{2}$. The surrogate is OU with $k_\nu = 2$, $\sigma_\nu = 1$, $\bar{x} = 0$; the temperature is $\lambda = 1$. With $\nabla\cdot b_\nu = -k_\nu$ and $a_\nu = 1 - k_\nu\,\delta t$, the discrete action and weight (20) are

$$S_{\nu,M}[\xi] \;=\; \frac{1}{2\,\delta t}\sum_{m=0}^{M-1}\big(\xi_{m+1} - a_\nu\,\xi_m\big)^2 \;-\; \frac{k_\nu T}{2}, \qquad w_\nu(\xi) \;=\; e^{-\lambda S_{\nu,M}[\xi]} \;\le\; W_{\max} = e^{\lambda k_\nu T/2} = e.$$

The observable is the quadratic terminal functional $F(\xi) = \xi_M^2 - 1$; it is unbounded, with all moments finite under the Gaussian measures below, so Theorem A and the probability bound of Theorem B–I apply exactly, while the bounded-observable statements apply to truncations $F \wedge R$.

9.2 Closed-form structure: the selected measure, the estimand, and the variance

Conditionally on $\xi_0 = 0$, both Π_M and the weight are Gaussian in $(\xi_1, \dots, \xi_M) \in \mathbb{R}^M$: $\Pi_M = \mathcal{N}(0, K_\Pi^{-1})$ and $w_\nu \propto \exp(-\frac{\lambda}{2\delta t}\sum_m(\xi_{m+1} - a_\nu\xi_m)^2)$, with K_Π and K_ν the tridiagonal precision matrices of the two quadratic forms. Hence the selected measure and the w^2-tilted measure are themselves Gaussian:

$$Q_{\nu,M} \;=\; \mathcal{N}\big(0,\ \Sigma_\lambda\big), \quad \Sigma_\lambda = (K_\Pi + \lambda K_\nu)^{-1}; \qquad \frac{w_\nu^2\,d\Pi_M}{\mathbb{E}[w_\nu^2]} \;=\; \mathcal{N}\big(0,\ \Sigma_{2\lambda}\big), \quad \Sigma_{2\lambda} = (K_\Pi + 2\lambda K_\nu)^{-1}.$$

Three exact quantities follow, none requiring any sampling.

The estimand. With $c := (\Sigma_\lambda)_{MM} = 0.158061$ the selected terminal variance,

$$F^*(\nu) \;=\; \mathbb{E}_{Q_{\nu,M}}[\xi_M^2] - 1 \;=\; c - 1 \;=\; -0.841939. \tag{22}$$

This exact evaluation is the benchmark for the convergence study below; no Monte Carlo benchmark is used.

The chi-square divergence. The divergence-correction constants cancel in the ratio and

$$\chi^2\big(Q_{\nu,M}\,\|\,\Pi_M\big) \;=\; \frac{\det(K_\Pi+\lambda K_\nu)}{\det(K_\Pi)^{1/2}\,\det(K_\Pi+2\lambda K_\nu)^{1/2}} \;-\; 1 \;=\; 1372.4, \tag{23}$$

with the Gaussian Kullback–Leibler divergence likewise available in closed form: $D_{\mathrm{KL}}(Q_{\nu,M}\|\Pi_M) = 4.860$, so $\exp(D_{\mathrm{KL}}) \approx 129.0$, consistent with Theorem 6.1(iii).

The asymptotic variance. By Theorem 6.1(i) and Gaussian fourth moments, with $s^2 := (\Sigma_{2\lambda})_{MM} = 0.096803$,

$$\sigma^2(\nu) \;=\; \frac{\mathbb{E}_{\Pi_M}[(F-F^*)^2 w_\nu^2]}{Z_\nu^2} \;=\; (1+\chi^2)\,\mathbb{E}_{\mathcal{N}(0,s^2)}\big[(\xi^2-c)^2\big] \;=\; (1+\chi^2)\big(3s^4-2c\,s^2+c^2\big) \;=\; 30.89, \tag{24}$$

so the CLT predictor is $\sqrt{\sigma^2(\nu)} \approx 5.56$. Note that (24) integrates the squared deviation against the w^2-tilted measure, not the selected measure: the incorrect factorization $(1+\chi^2)\,\mathrm{Var}_{Q_{\nu,M}}(F)$ (Remark 4.2) would give $(1+\chi^2)\cdot 2c^2 = 68.6$ here, more than twice the true value. Finally $Z_\nu = 7.87\times 10^{-8}$, so $W_{\max}/Z_\nu \approx 3.5\times 10^7$: the weights are bounded (Tier II applies in principle) but the bounded-weight exponents of (16)–(17) are vacuous at practical N, and the chi-square-based Tier I and the diagnostics of Theorem C carry the operational content, exactly as Remark 5.4 anticipates.

9.3 Numerical convergence study

We compute $\hat{F}_N(\nu)$ for $N = 10^2, 3\times 10^2, 10^3, 3\times 10^3, 10^4, 3\times 10^4, 10^5$ over 50 independent replications and compare against the exact benchmark (22). Table 1 reports the absolute empirical bias, the standard deviation, the $\sqrt{N}$-scaled standard deviation, and the median effective sample size $\hat{N}_{\mathrm{eff}}$.

N	|bias|	std($\hat{F}_N$)	$\sqrt{N}\cdot$ std	median $\hat{N}_{\mathrm{eff}}$
100	5.51×10^{-2}	1.97×10^{-1}	1.97	2.7
300	2.02×10^{-2}	1.12×10^{-1}	1.94	4.8
1 000	1.58×10^{-2}	7.68×10^{-2}	2.43	7.4
3 000	2.36×10^{-2}	7.36×10^{-2}	4.03	12.4
10 000	8.48×10^{-3}	4.75×10^{-2}	4.75	26.5
30 000	1.88×10^{-4}	2.47×10^{-2}	4.28	60.0
100 000	6.42×10^{-4}	1.51×10^{-2}	4.77	136.7

Table 1: Convergence of $\hat{F}_N(\nu)$ for the worked OU example over 50 replications, against the *exact* benchmark $F^*(\nu) = -0.841939$ of (22). The $\sqrt{N}\cdot$ std column rises, non-monotonically at 50 replications, toward the exact CLT asymptote $\sqrt{\sigma^2(\nu)} \approx 5.56$ of (24) and is still $\approx 14\%$ below it at $N = 10^5$: the empirical variance, like the plug-in $\hat{\chi}^2$ of §9.4, is dominated by rare large weights and is negatively biased at finite N. Median $\hat{N}_{\mathrm{eff}}/N \approx 1.4\times 10^{-3}$ at $N = 10^5$, above its almost-sure limit $1/(1+\chi^2) = 7.3\times 10^{-4}$ (Theorem 6.1(ii)) for the same reason.

9.4 Effective sample size and the two scales

Evaluating the closed forms exactly gives

$$\chi^2\big(Q_{\nu,M}\,\|\,\Pi_M\big) \;=\; 1372.4, \qquad D_{\mathrm{KL}}\big(Q_{\nu,M}\,\|\,\Pi_M\big) \;=\; 4.860.$$

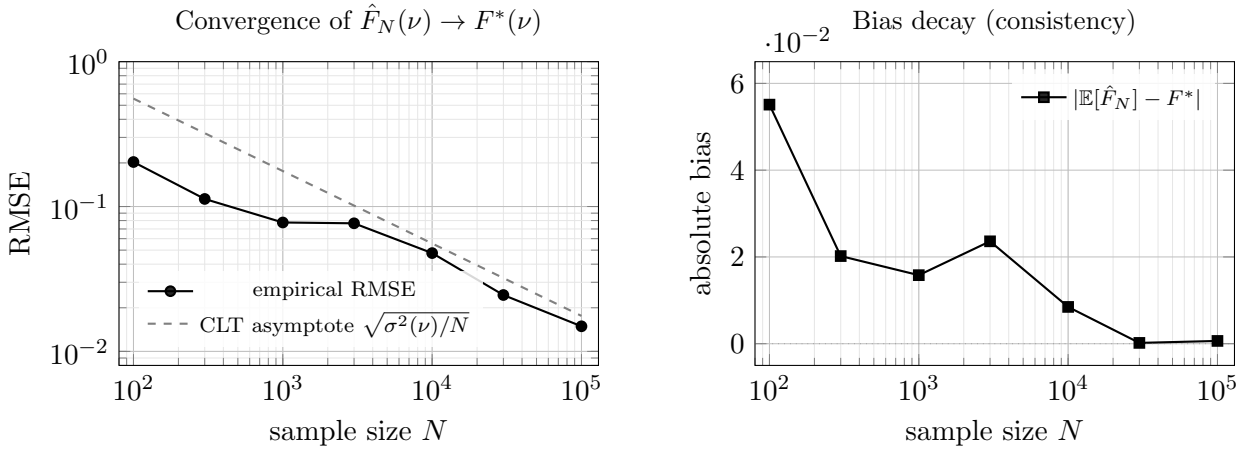


Figure 1: Numerical convergence of the SNIS estimator $\hat{F}_N(\nu)$ for the worked one-dimensional OU example, against the exact benchmark $F^*(\nu)$ of (22). Left: empirical RMSE versus N (log–log) with the exact CLT reference $\sqrt{\sigma^2(\nu)/N}$ from (24); the empirical curve approaches the asymptote from below as the weight tail is progressively resolved. Right: absolute empirical bias versus N (log–linear). Bias decay illustrates Theorem 4.1(i); the RMSE scaling illustrates Theorem 4.1(ii).

The Chatterjee–Diaconis scale is $\exp(D_{\mathrm{KL}}) \approx 129$; the Sanz-Alonso scale is $1 + \chi^2 \approx 1373$; the example lands in the chi-square-binding regime anticipated in §6, the two scales differing by an order of magnitude. A plug-in Monte Carlo estimate of (23) from an $N = 2 \times 10^5$ proposal sample reads only $\hat{\chi}^2 \approx 623$, less than half the exact value: the plug-in estimator is itself dominated by the heavy upper tail of w_ν and severely negatively biased at this sample size, and the median $\hat{N}_{\mathrm{eff}}/N \approx 1.4 \times 10^{-3}$ of Table 1 is the same biased diagnostic (its almost-sure limit is $1/(1+\chi^2) = 7.3 \times 10^{-4}$). The gap is itself instructive: the chi-square diagnostic that sets the binding sample-size scale cannot be read reliably off a moderate proposal sample in precisely the heavy-tailed regime for which the reweighting object is needed, so the closed form (23) is not a convenience but a necessity. The empirical convergence in Table 1 is consistent with the exact scales: at $N = 10^2 \approx \exp(D_{\mathrm{KL}})$ the estimator carries std ≈ 0.20, while stabilization to std $\lesssim 0.05$ occurs for $N \gtrsim 10^4 \approx 7 \times (1 + \chi^2)$.

Grid refinement and the continuum degeneracy, quantitatively Proposition 2.3 asserts that the continuous-path weights vanish almost surely; the closed form (23) shows what that degeneracy looks like from the grid. With $\lambda = 1$ and the physical parameters held fixed, refining the grid gives

M	25	50	100	200
$1 + \chi^2$	3.77×10^1	1.37×10^3	1.82×10^6	3.22×10^{12}
Z_ν	4.6×10^{-4}	7.9×10^{-8}	2.3×10^{-15}	2.1×10^{-30}

so the normalizing constant collapses and the chi-square divergence explodes, exponentially in M (at rate $1 + \chi^2 \approx e^{0.144\,M}$ for these parameters): the discrete shadow of the continuum degeneracy. At any fixed M every quantity is finite and Theorems A–C apply verbatim; the grid is the model, and any continuum limit requires a renormalized temperature or action (§11).

9.5 Intellectual-property discipline

The example uses textbook quantities: a discretized one-dimensional OU proposal released from the origin, a one-dimensional OU surrogate at a different rate, a Gaussian-quadratic discrete action, a quadratic terminal observable, and a forward Euler–Maruyama discretizer. No details of any production algorithm are present. All quantities in (22)–(24), Table 1, and Figure 1 are reproducible from the closed-form expressions and the discretizer stated above.

10 Structural comparison with classical analogues

We compare the selection-functional construction with three classical frameworks: misspecified-model theory, path-space variational inference, and generalized Bayes / PAC-Bayes. The comparison is summarized in Table 2 and organized around two structural distinctions.

10.1 Misspecified-model theory

White's analysis of the quasi-maximum-likelihood estimator under misspecification [75] establishes that, under regularity conditions, $\hat{\theta}_n$ converges to the pseudo-true value $\theta_* = \arg\min_\theta D_{\mathrm{KL}}(P_0 \,\|\, P_\theta)$, where P_0 is the true law and $\{P_\theta\}$ the fitted family, with sandwich-covariance asymptotic normality [37, 53]. The Bayesian analogue [41, 42, 73, 31] establishes Bernstein–von Mises theorems under misspecification, with the posterior concentrating at θ_* but with model-based rather than sandwich covariance [53]. The structural assumption throughout is that the true law P_0 exists as a well-defined object against which the family is scored in Kullback–Leibler (KL) divergence. In the selection-functional setting, on the observed state description there is no generator-based P_0 to score against (Proposition 2.1), and the surrogate family cannot even reproduce the target's two-time kernels uniformly (Proposition 7.3). The calibrated ν^* of §8.4 is well defined as the population minimizer of the surrogate action, but it is the best reweighting parameter in the chosen family, not the Kullback–Leibler projection of a generator-based truth.

10.2 Path-space variational inference

Variational inference [72, 8] targets an intractable but well-defined posterior by minimizing a divergence over a tractable family; importance-weighted variants [10, 49, 54, 21] tighten the bound with multi-sample estimators, and chi-square [20] and Rényi [45] divergences extend the divergence family. Path-space instances include Schrödinger-bridge formulations [44, 17, 57], control-based importance sampling of diffusions [33, 76], stochastic-optimal-control matching [35], and the path-space variational coarse-graining of [32], which is the closest published prior art: it projects a non-Markovian molecular dynamics onto a Markovian surrogate by minimizing a path-space relative entropy. Two distinctions separate that construction from the present one. (a) It presupposes a well-defined path-space relative entropy between the true dynamics and the surrogate family—a generator-derived density evaluable along paths; under indivisibility no such density exists on the observed description (Remark 3.8). (b) It optimizes the surrogate toward the truth, whereas the present construction holds the surrogate fixed as a reweighting object, and Proposition 7.3 bounds what optimization toward the target's kernels could achieve in any case. The selection-functional construction is therefore not a path-space variational projection; it is an SNIS estimator whose target is the selected measure itself.

10.3 Generalized Bayes and PAC-Bayes

The Bissiri–Holmes–Walker framework [7] characterizes the coherent update of a prior Π by a loss ℓ as $p^{(\mathrm{GB})} \propto e^{-\lambda R}\,\Pi$, with R the cumulative loss; the selected measure (9) is the path-space instance with $\ell = S_{\nu,M}$ acting on the candidate path itself (Remark 3.8). PAC-Bayes risk bounds [12, 2, 3, 43] for Gibbs posteriors $\hat{\rho}_\lambda \propto e^{-\lambda r}\pi$, with r the empirical risk, transfer to the calibrated surrogate level (§8.4): they bound the generalization of $\hat{\nu}$ against the population surrogate risk. They do not translate into bounds against the target process: that translation is exactly hypothesis (D1) of Theorem 7.1, and it is left as the assumption it is.

10.4 Two structural distinctions

	Misspecified-model theory	Variational inference	Selection functional (this paper)
Assumed object	True P_0, scored in KL	Well-defined posterior $p(\theta \mid y)$	Two-time kernels $\Gamma(t \leftarrow s)$ only; no generator on the observed space
Fitting target	KL projection θ_*	KL-closest q^*_ϕ in tractable family	Selected measure $Q_{\nu,M}$; estimand $F^*(\nu)$
Convergence object	$\hat{\theta}_n \to \theta_*$, sandwich CLT	$q^*_\phi \to p$ in chosen divergence	$\hat{F}_N(\nu) \to F^*(\nu)$, CLT (12)
Truth link	Definitional (P_0 assumed)	Definitional (posterior assumed)	Conditional transfer under (D1); obstruction (19)

Table 2: Structural comparison. Two distinctions organize the column on the right and hold under indivisibility but in neither classical analogue: (a) no generator on the observed state space reproduces the target's two-time family, so there is no generator-based true law to score against or project onto; (b) no generator-derived likelihood or posterior exists on path space over the observed description, so there is nothing for a variational method to approximate. Both distinctions are statements about the observed description (Remark 2.2); neither denies the existence of hidden-variable enlargements or of path laws for process-induced targets.

The two distinctions in the caption of Table 2 are exactly the corrected exclusion (E1) of Remark 2.2 and its likelihood-level consequence (Remark 3.8). We record explicitly that an earlier version of this paper claimed a third distinction—a local-in-time scope for the surrogate action—which does not withstand scrutiny: the action (20) integrates over the full horizon, and the restoring rate produces an effective decay of sensitivity, not locality of the functional (§8.1). The comparison rests on the two distinctions that survive.

11 Discussion and open problems

We have developed the OM action in a third operational role, distinct from the path-density and rate-functional roles: as a selection functional acting on externally proposed grid paths, computed against a surrogate that is explicitly not the truth. The theory delivered is exactly the theory claimed: strong consistency, asymptotic normality, and asymptotic mean-squared error for the estimator of the selected-measure expectation (Theorem A); finite-sample concentration in two tiers

matched to the weight regime (Theorem B); variance and effective-sample-size diagnostics organized by the chi-square divergence, with the published sample-size results placed as diagnostic scales under their hypotheses (Theorem C); and a truth link stated as what it is—a conditional transfer under an explicit measure-level hypothesis, flanked by an unconditional obstruction bounding every divisible surrogate away from an indivisible target (Theorem D, Proposition 7.3). The worked example evaluates every one of these objects in closed form and measures its estimator against an exact benchmark.

Five problems are left open.

The truth-link bridge Control the tolerance $\eta_{\nu,M}$ of hypothesis (D1) by indivisibility invariants of the target—the integrated deficit $J(\mathcal{R})$ of [65, Theorem 7.4] or the liftability profile h_d and memory rank of [67]—for principled proposal–surrogate pairs. This is the selection bridge of [67, Open Problem (i)], stated there from the geometric side and here from the estimation side; it is the single question on which the two papers' programs meet, and it is open (Remark 7.2).

Continuous-path theory Construct a renormalized continuous-path selection theory to which the grid theory converges: candidate routes are tube-limit renormalizations of the OM functional [22, 11], Cameron–Martin shifts of the proposal, and Γ-convergence of the discrete actions [4, 60] with a temperature schedule $\lambda_M \downarrow$ calibrated to the blow-up quantified in §9.4. Proposition 2.3 shows the naive limit (λ fixed) does not exist; the refinement table shows the rate at which it fails.

Optimal surrogate Characterize $\nu^\dagger = \arg\min_\nu \sigma^2(\nu)$ for a fixed observable class—an unconditional, variance-only optimization for which Theorem C supplies the governing quantities—and, under hypothesis (D1) with known tolerances $\eta_{\nu,M}$, the full risk trade-off $\eta_{\nu,M}^2 + \sigma^2(\nu)/N$. For the OU surrogate the sweep is one-dimensional in the restoring rate k, with the chi-square scale of Theorem C bounding the useful range of k from above.

Adaptive proposals Allow the proposal Π_n to depend on the previous $n-1$ samples (sequential importance sampling or particle filtering, exploiting the causal factorization of Remark 3.9). Consistency extends along standard lines; the CLT requires martingale central limit machinery in the spirit of [18, 15].

Multilevel selection Develop a multilevel Monte Carlo analogue [27] of the estimator across a hierarchy of grids $M_0 < M_1 < \cdots$, coupling levels to offset the growth of the per-level chi-square divergence quantified in §9.4. Whether coupled increments can be arranged so that the level-wise variances decay while the level-wise χ^2 divergences grow is the open design question.

Declarations

AI assistance was used in the preparation of this manuscript. All theoretical frameworks, methodological approaches, and conclusions are the author's original work; the author reviewed all content and assumes responsibility for it.

Competing interests

The author declares no competing interests. Funding information appears in the footnote to the title.

Data and code availability

The worked example of §9 is fully specified in the text: the proposal, surrogate, discrete action, observable, discretizer, and grid are given explicitly, the benchmark and diagnostic quantities are closed-form functions of the stated precision matrices, and the Monte Carlo study is reproducible from the stated specification. No experimental data were generated or analyzed.

References

[1] S. Agapiou, O. Papaspiliopoulos, D. Sanz-Alonso, and A. M. Stuart, Importance sampling: intrinsic dimension and computational cost. *Statistical Science* **32**(3) (2017), 405–431.

[2] P. Alquier, User-friendly introduction to PAC-Bayes bounds. *Foundations and Trends in Machine Learning* **17**(2) (2024), 174–303.

[3] P. Alquier, J. Ridgway, and N. Chopin, On the properties of variational approximations of Gibbs posteriors. *Journal of Machine Learning Research* **17**(239) (2016), 1–41.

[4] B. Ayanbayev, I. Klebanov, H. C. Lie, and T. J. Sullivan, Γ-convergence of Onsager–Machlup functionals: I and II. *Inverse Problems* **38** (2022), 025005 and 025006.

[5] J. A. Barandes, The stochastic-quantum correspondence. *Philosophy of Physics* **3**(1) (2025), 186.

[6] J. A. Barandes, Quantum systems as indivisible stochastic processes. arXiv:2507.21192 (2025).

[7] P. G. Bissiri, C. C. Holmes, and S. G. Walker, A general framework for updating belief distributions. *Journal of the Royal Statistical Society Series B* **78**(5) (2016), 1103–1130.

[8] D. M. Blei, A. Kucukelbir, and J. D. McAuliffe, Variational inference: a review for statisticians. *Journal of the American Statistical Association* **112**(518) (2017), 859–877.

[9] S. Boucheron, G. Lugosi, and P. Massart, *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

[10] Y. Burda, R. Grosse, and R. Salakhutdinov, Importance weighted autoencoders. *International Conference on Learning Representations* (2016). arXiv:1509.00519.

[11] M. Capitaine, Onsager–Machlup functional for some smooth norms on Wiener space. *Probability Theory and Related Fields* **102** (1995), 189–201.

[12] O. Catoni, *PAC-Bayesian Supervised Classification: The Thermodynamics of Statistical Learning*. IMS Lecture Notes–Monograph Series, Vol. 56, 2007.

[13] Y. Chao and J. Duan, The Onsager–Machlup function as Lagrangian for the most probable path of a jump-diffusion process. arXiv:1812.06409 (2019).

[14] S. Chatterjee and P. Diaconis, The sample size required in importance sampling. *Annals of Applied Probability* **28**(2) (2018), 1099–1135.

[15] N. Chopin and O. Papaspiliopoulos, *An Introduction to Sequential Monte Carlo*. Springer, 2020.

[16] A. J. Chorin, O. H. Hald, and R. Kupferman, Optimal prediction with memory. *Physica D* **166**(3–4) (2002), 239–257.

[17] V. De Bortoli, J. Thornton, J. Heng, and A. Doucet, Diffusion Schrödinger bridge with applications to score-based generative modeling. *NeurIPS* **34** (2021). arXiv:2106.01357.

[18] P. Del Moral, *Feynman–Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, 2004.

[19] A. Dembo and O. Zeitouni, *Large Deviations Techniques and Applications*, 2nd ed. Springer, 1998.

[20] A. B. Dieng, D. Tran, R. Ranganath, J. Paisley, and D. M. Blei, Variational inference via χ upper bound minimization. *NIPS* (2017).

[21] J. Domke and D. Sheldon, Importance weighting and variational inference. *NeurIPS* **31** (2018), 4471–4480.

[22] D. Dürr and A. Bach, The Onsager–Machlup function as Lagrangian for the most probable path of a diffusion process. *Communications in Mathematical Physics* **60** (1978), 153–170.

[23] W. E, W. Ren, and E. Vanden-Eijnden, Minimum action method for the study of rare events. *Communications on Pure and Applied Mathematics* **57** (2004), 637–656.

[24] W. Feller, Non-Markovian processes with the semigroup property. *Annals of Mathematical Statistics* **30**(4) (1959), 1252–1253.

[25] M. I. Freidlin and A. D. Wentzell, *Random Perturbations of Dynamical Systems*, 3rd ed. Springer, 2012.

[26] J. Geweke, Bayesian inference in econometric models using Monte Carlo integration. *Econometrica* **57**(6) (1989), 1317–1339.

[27] M. B. Giles, Multilevel Monte Carlo path simulation. *Operations Research* **56**(3) (2008), 607–617.

[28] D. Givon, R. Kupferman, and A. Stuart, Extracting macroscopic dynamics: model problems and algorithms. *Nonlinearity* **17**(6) (2004), R55–R127.

[29] P. W. Glynn and W. Whitt, The asymptotic efficiency of simulation estimators. *Operations Research* **40**(3) (1992), 505–520.

[30] T. Grafke, T. Schäfer, and E. Vanden-Eijnden, Sharp asymptotic estimates for expectations, probabilities, and mean first passage times in stochastic systems with small noise. arXiv:2103.04837 (2021).

[31] P. Grünwald and T. van Ommen, Inconsistency of Bayesian inference for misspecified linear models, and a proposal for repairing it. *Bayesian Analysis* **12**(4) (2017), 1069–1103.

[32] V. Harmandaris, E. Kalligiannaki, M. Katsoulakis, and P. Plecháč, Path-space variational inference for non-equilibrium coarse-grained systems. *Journal of Computational Physics* **314** (2016), 355–383.

[33] C. Hartmann and C. Schütte, Efficient rare event simulation by optimal nonequilibrium forcing. *Journal of Statistical Mechanics: Theory and Experiment* (2012), P11004.

[34] M. Heymann and E. Vanden-Eijnden, The geometric minimum action method: a least action principle on the space of curves. *Communications on Pure and Applied Mathematics* **61** (2008), 1052–1117.

[35] L. Holdijk, Y. Du, F. Hooft, P. Jaini, B. Ensing, and M. Welling, Stochastic optimal control matching for transition path sampling. arXiv:2410.07974 (2024).

[36] C. C. Holmes and S. G. Walker, Assigning a value to a power likelihood in a general Bayesian model. *Biometrika* **104**(2) (2017), 497–503.

[37] P. J. Huber, The behavior of maximum likelihood estimates under nonstandard conditions. *Proc. Fifth Berkeley Symp. Math. Statist. and Probability* **1** (1967), 221–233.

[38] J. Hu and J. Duan, Onsager–Machlup action functional for stochastic partial differential equations with Lévy noise. arXiv:2011.09690 (2020).

[39] J. Huang, X. Zhou, and J. Duan, Probability flow approach to the Onsager–Machlup functional for jump-diffusion processes. arXiv:2409.01340 (2024).

[40] N. Ikeda and S. Watanabe, *Stochastic Differential Equations and Diffusion Processes*, 2nd ed. North-Holland, 1989.

[41] B. J. K. Kleijn and A. W. van der Vaart, Misspecification in infinite-dimensional Bayesian statistics. *Annals of Statistics* **34**(2) (2006), 837–877.

[42] B. J. K. Kleijn and A. W. van der Vaart, The Bernstein–von Mises theorem under misspecification. *Electronic Journal of Statistics* **6** (2012), 354–381.

[43] J. Knoblauch, J. Jewson, and T. Damoulas, An optimization-centric view on Bayes' rule: reviewing and generalizing variational inference. *Journal of Machine Learning Research* **23**(132) (2022), 1–109.

[44] C. Léonard, A survey of the Schrödinger problem and some of its connections with optimal transport. *Discrete and Continuous Dynamical Systems* **34**(4) (2014), 1533–1574.

[45] Y. Li and R. E. Turner, Rényi divergence variational inference. *NIPS* (2016).

[46] K. K. Lin and F. Lu, Data-driven model reduction, Wiener projections, and the Koopman–Mori–Zwanzig formalism. *Journal of Computational Physics* **424** (2021), 109864.

[47] Y. T. Lin, Y. Tian, D. Perez, and D. Livescu, Regression-based projection for learning Mori–Zwanzig operators. *SIAM Journal on Applied Dynamical Systems* **22**(4) (2023), 2890–2926.

[48] J. S. Liu, Metropolized independent sampling with comparisons to rejection sampling and importance sampling. *Statistics and Computing* **6** (1996), 113–119.

[49] C. J. Maddison, J. Lawson, G. Tucker, N. Heess, M. Norouzi, A. Mnih, A. Doucet, and Y. W. Teh, Filtering variational objectives. *NIPS* (2017).

[50] S. Milz and K. Modi, Quantum stochastic processes and quantum non-Markovian phenomena. *PRX Quantum* **2** (2021), 030201.

[51] H. Mori, Transport, collective motion, and Brownian motion. *Progress of Theoretical Physics* **33** (1965), 423–455.

[52] P. Mörters and Y. Peres, *Brownian Motion*. Cambridge University Press, 2010.

[53] U. K. Müller, Risk of Bayesian inference in misspecified models, and the sandwich covariance matrix. *Econometrica* **81**(5) (2013), 1805–1849.

[54] C. A. Naesseth, S. W. Linderman, R. Ranganath, and D. M. Blei, Variational sequential Monte Carlo. *AISTATS* (2018).

[55] F. Nüske, B. G. Keller, G. Pérez-Hernández, A. S. J. S. Mey, and F. Noé, Variational approach to molecular kinetics. *Journal of Chemical Theory and Computation* **10**(4) (2014), 1739–1752.

[56] A. B. Owen, *Monte Carlo Theory, Methods and Examples*. Online textbook, Stanford University, 2013.

[57] M. Pavon, E. G. Tabak, and G. Trigila, The data-driven Schrödinger bridge. *Communications on Pure and Applied Mathematics* **74** (2021), 1545–1573.

[58] V. H. de la Peña, T. L. Lai, and Q.-M. Shao, *Self-Normalized Processes: Limit Theory and Statistical Applications*. Springer, 2009.

[59] F. J. Pinski and A. M. Stuart, Transition paths in molecules at finite temperature. *Journal of Chemical Physics* **132** (2010), 184104.

[60] F. J. Pinski, A. M. Stuart, and F. Theil, Γ-limit for transition paths of maximal probability. *Journal of Statistical Physics* **146** (2012), 955–974.

[61] D. Sanz-Alonso, Importance sampling and necessary sample size: an information theory approach. *SIAM/ASA Journal on Uncertainty Quantification* **6**(2) (2018), 867–879.

[62] Z. Selk and H. Honnappa, The small-noise limit of the most likely element is the most likely element in the small-noise limit. arXiv:2209.04523 (2022).

[63] P. Stinis, Renormalized Mori–Zwanzig-reduced models for systems without scale separation. *Proceedings of the Royal Society A* **471** (2015), 20140446.

[64] B. D. Summers, The incompleteness of the instantaneous state and the status of Chapman–Kolmogorov divisibility: analysis from limits, relativity, and renormalization. SSRN, `https://doi.org/10.2139/ssrn.6922380` (2026).

[65] B. D. Summers, Koopman spectral theory for indivisible stochastic processes. SSRN, `https://doi.org/10.2139/ssrn.6733181` (2026).

[66] B. D. Summers, Convex perturbation and the suppression of the indivisibility deficit. SSRN, `https://doi.org/10.2139/ssrn.6812461` (2026).

[67] B. D. Summers, Memory rank and liftability of transition families without a generator: a projection theory for the indivisibility deficit. SSRN preprint (2026).

[68] B. D. Summers, Memory production under coarse-graining: a balance law for the indivisibility deficit. SSRN preprint (2026).

[69] M. Uchida and N. Yoshida, Estimation for misspecified ergodic diffusion processes from discrete observations. *ESAIM: Probability and Statistics* **15** (2011), 270–290.

[70] T. van Erven and P. Harremoës, Rényi divergence and Kullback–Leibler divergence. *IEEE Transactions on Information Theory* **60**(7) (2014), 3797–3820.

[71] A. Vehtari, D. Simpson, A. Gelman, Y. Yao, and J. Gabry, Pareto smoothed importance sampling. *Journal of Machine Learning Research* **25**(72) (2024), 1–58.

[72] M. J. Wainwright and M. I. Jordan, Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning* **1**(1–2) (2008), 1–305.

[73] S. G. Walker, Bayesian inference with misspecified models. *Journal of Statistical Planning and Inference* **143**(10) (2013), 1621–1633.

[74] Y. Wang, Z. Liu, and J. Duan, Onsager–Machlup functional for stochastic differential equations with time-varying noise. arXiv:2407.04290 (2024).

[75] H. White, Maximum likelihood estimation of misspecified models. *Econometrica* **50**(1) (1982), 1–25.

[76] W. Zhang, H. Wang, C. Hartmann, M. Weber, and C. Schütte, Applications of the cross-entropy method to importance sampling and optimal control of diffusions. *SIAM Journal on Scientific Computing* **36**(6) (2014), A2654–A2672.

[77] R. Zwanzig, Ensemble method in the theory of irreversibility. *Journal of Chemical Physics* **33** (1960), 1338–1341.

[78] R. Zwanzig, Memory effects in irreversible thermodynamics. *Physical Review* **124** (1961), 983–992.